Article

PIR-SBFR: Physical Imaging Reliability-Guided Scale-Biased Feature Reweighting for Robust Object Detection in Optical Remote Sensing Imagery

Zizheng Zhao [1], Jingchao Liu [1,*], Zixin Wang [1], Xiaoyu Dong [1], Zhirui Xue [1], Junhao Hu [1] and Chengxin Zhu [1]

[1] Xijing University, No. 1 Xijing Road, Chang'an District, Xi'an, Shaanxi Province, China; 2508540402019@stu.xijing.edu.cn
* Correspondence: 20100058@xijing.edu.cn

Highlights

What are the main findings?

- PIR-SBFR separates the pyramid levels demanded by scene content from the levels that remain reliable under the observed acquisition condition.
- Across three seeds, AP improves by 2.54 points on DIOR and 2.52 points on AI-TOD-v2; all 27 controlled test cells and nine unseen degradations improve (unseen mean: +4.37 points).

What are the implications of the main findings?

- Separate mixed-degradation controls show that correctly paired descriptors improve routing without becoming a single point of failure.
- The P5 bypass restores coarse structural evidence and the deployed router adds 4.63% GFLOPs, although category-wise gains are not uniform.

Abstract

Multiscale fusion in optical remote-sensing detectors typically ignores how acquisition conditions change pyramid reliability. A small-object scene may require P3 features, while coarse ground sampling distance (GSD), optical blur characterized by the modulation transfer function or point-spread function (MTF/PSF), or low signal-to-noise ratio (SNR) weakens them. We propose Physical Imaging Reliability-Guided Scale-Biased Feature Reweighting (PIR-SBFR), an acquisition-conditioned extension of YOLO-DSF that separates scene-level scale demand from imaging-dependent reliability. An analytic branch maps GSD, MTF/PSF, SNR, and availability masks to a monotonic P3–P5 reliability prior; a scale-supervised visual branch supplies a residual demand correction, and logit-space fusion produces normalized routing weights. Metadata dropout, neutral-prior fallback, degradation-consistency regularization, and a P5 structural bypass support incomplete records and preserve coarse evidence. Across three seeds on DIOR, PIR-SBFR improves COCO-style AP from 62.98% to 65.52% over YOLO-DSF and small-object AP by 3.53 percentage points. AP also improves by 2.52 points on AI-TOD-v2. PIR-SBFR improves all 27 GSD–MTF–SNR test cells and gains 4.37 points across nine unseen degradations, while adding 4.63% GFLOPs. These results support robust and interpretable acquisition-conditioned multiscale fusion.

Keywords: Optical Remote Sensing; Aerial Object Detection; Physical Imaging Reliability; Acquisition Metadata; Multiscale Feature Routing; Small Object Detection; YOLO

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1. Introduction

Optical remote-sensing imagery provides wide-area, repeated observations for land-resource management, urban and traffic monitoring, maritime surveillance, precision agriculture, and disaster response[1–3]. Object detectors convert these observations into operational information about vehicles, ships, aircraft, buildings, transport facilities, and other man-made structures. Unlike many ground-level natural images, however, an overhead image can contain categories whose pixel occupancies differ by orders of magnitude. Small vehicles and ships may cover only a few pixels, whereas bridges, dams, and airports require much broader spatial context. Dense layouts, background textures, shadows, and weak object boundaries further complicate the separation of target evidence from clutter[4–8]. Effective remote-sensing detection depends on extracting features at several resolutions and identifying which resolutions remain informative for each scene.

That decision is coupled to the image-formation process. Ground sampling distance (GSD) determines how physical extent is converted into pixels; the modulation transfer function or point-spread function (MTF/PSF) controls the attenuation of spatial detail; and signal-to-noise ratio (SNR) affects whether low-contrast target responses remain distinguishable from the background. Consequently, the nominal resolution of a feature-pyramid level does not guarantee that its information is reliable. A scene containing many small objects may have strong demand for the fine P3 level, while coarse sampling, blur, or noise simultaneously reduces the usable evidence at that level. Conversely, favoring coarse levels solely because the acquisition is degraded may suppress the limited fine detail still required to localize small targets. We refer to this distinction as the mismatch between *scene-level scale demand* and *acquisition-dependent feature reliability*.

Modern one-stage detectors, particularly the YOLO family, offer an effective basis for optical remote-sensing detection because they combine dense prediction with practical inference cost[4,5,9]. Feature Pyramid Networks (FPNs) and extensions such as BiFPN, ASFF, and AFPN improve cross-level aggregation through lateral connections, learned weighting, or adaptive spatial fusion[10–13]. Acquisition-conditioned detectors further show that GSD and platform metadata can contribute useful information, while FiLM provides a general mechanism for feature-wise conditioning[14–16]. Nevertheless, most multiscale routers infer scale weights primarily from visual activations, whereas generic metadata encoders do not ensure that their outputs represent level-wise physical reliability. In both cases, the reason for changing a P3–P5 weight is difficult to identify: the change may reflect the objects present, degradation of the observation, or an arbitrary correlation in the conditioning variables. Dependence on metadata also creates a deployment risk when acquisition fields are noisy, incomplete, or unavailable. A robust router should answer two separate questions: which scales the scene requires and how trustworthy those scales are under the current imaging condition.

This study develops Physical Imaging Reliability-Guided Scale-Biased Feature Reweighting (PIR-SBFR) to provide that separation. PIR-SBFR is built on YOLO-DSF, whose Dilated Receptive Field Block (DRFB) and Feature-Aware Coupled Head (FACH) preserve spatial evidence and coordinate classification with localization. The original visual-only SBFR neck is replaced by two complementary routing paths. An analytic path converts GSD, MTF/PSF, SNR, and descriptor-availability masks into a monotonic reliability prior over P3, P4, and P5. A visual path is supervised by the ground-truth small-, medium-, and large-object distribution during training and learns a residual correction for scene-dependent scale demand. The two signals are fused in logit space to obtain normalized routing weights. Metadata dropout and a neutral-prior fallback prevent failure when descriptors are missing, degradation-consistency regularization discourages unstable routing, and an unweighted P5 structural bypass preserves coarse evidence for large continuous objects.

The central research question is whether physically constrained, visually corrected routing improves multiscale detection by capturing the intended demand–reliability relationship instead of benefiting only from extra parameters or metadata. We address this question with complementary evidence on DIOR and AI-TOD-v2. The evaluation includes controlled component and fusion comparisons, a separate mixed-degradation metadata-control set, routing probes, a complete 27-cell GSD–MTF–SNR grid, nine held-out degradation operators, and an auxiliary recorded-metadata flight analysis. Across three seeds, PIR-SBFR improves DIOR AP by 2.54 percentage points and AI-TOD-v2 AP by 2.52 points over YOLO-DSF. Its mean gain across the nine unseen degradation conditions is 4.37 points, while the deployed model adds 4.63% GFLOPs. These experiments test predictive performance and whether the routing behavior follows the proposed physical interpretation.

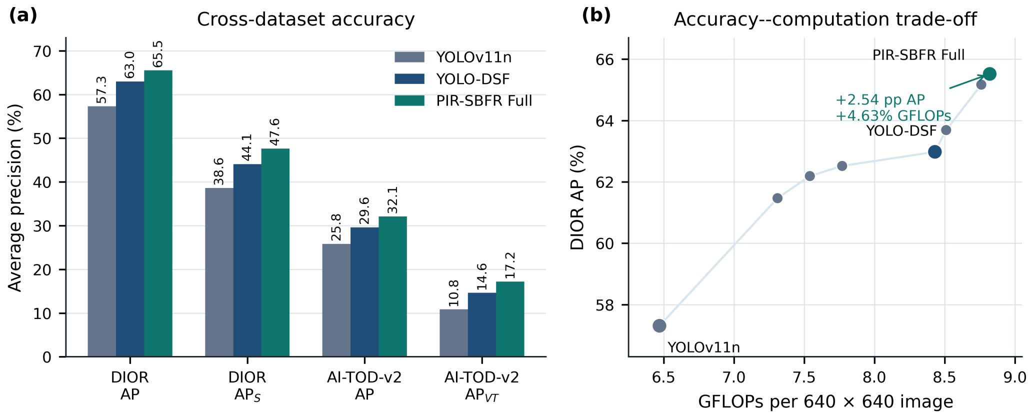

Figure 1. Overview of the principal performance and efficiency results. (a) AP and small/tiny-object AP on DIOR and AI-TOD-v2. (b) DIOR AP versus GFLOPs for all efficiency-tested variants. Values are three-seed means; the efficiency test uses FP16 and batch size 1. Detailed protocols and complete results are reported in Sections 3 and 4, respectively.

The main contributions of this work are as follows:

(1) We formulate cross-scale fusion as the combination of scene-level scale demand and acquisition-dependent feature reliability. GSD, MTF/PSF, SNR, and availability masks are mapped to an explicit, monotonic P3–P5 prior instead of being treated as an unconstrained metadata embedding.

(2) We introduce a scale-supervised visual residual and logit-space fusion mechanism that corrects the analytic prior when scene content and nominal physical reliability disagree. The resulting weights remain normalized and directly inspectable at each pyramid level.

(3) We incorporate metadata dropout, neutral-prior fallback, degradation-consistency regularization, and a P5 structural bypass so that the detector can use acquisition information without requiring complete metadata or discarding coarse structural evidence.

(4) We evaluate the mechanism through cross-dataset accuracy, controlled alternatives, separate mixed-degradation metadata controls, routing analysis, factorial and unseen-degradation tests, efficiency measurements, and auxiliary flight data. The evidence supports the performance gain and the intended interpretation of the router while identifying its remaining category- and telemetry-related limitations.

2. Related Work

2.1. Optical Remote-Sensing and Small-Object Detection

The development of optical remote-sensing detection has shifted from handcrafted intensity, texture, and shape descriptors toward learned hierarchical representations. Surveys and the DIOR benchmark identify large scale variation, dense spatial layouts, arbitrary viewing geometry, and background confusion as persistent sources of error[1,2]. These difficulties are especially pronounced for tiny and small objects because repeated downsampling can remove the few pixels that distinguish a target from roofs, roads, vegetation, water, or shadows. Local texture is critical for vehicles and ships, whereas facilities such as bridges, dams, and stadiums depend more strongly on extended structure.

Recent YOLO-based remote-sensing detectors pursue several complementary strategies. RSI-YOLO combines channel–spatial attention, weighted bidirectional pyramid fusion, and an additional small-object head[4]. MSA-YOLO uses multiscale strip attention to suppress background interference while retaining objects of different extents[5]. LI-YOLO couples low-illumination feature enhancement with adaptive spatial feature fusion[6]. YOLO-DroneMS and HSP-YOLOv8 further emphasize multiscale processing for dense UAV targets[7,8]. These methods establish the value of detail preservation, context, and adaptive fusion, but they generally treat the image as the complete description of the observation. The acquisition process is not represented as an explicit reliability constraint on each pyramid level.

2.2. Multiscale Representation and Feature-Pyramid Fusion

Feature pyramids provide high-resolution maps for small objects and progressively coarser semantic maps for larger structures. FPN builds semantically strong multiscale maps through a top-down path and lateral connections[10]. BiFPN introduces repeated bidirectional fusion with learned normalized weights[11]; ASFF performs spatially adaptive combination of features from different scales[12]; and AFPN progressively incorporates higher-level features to reduce semantic gaps[13]. Remote-sensing detectors also use weighted pyramid fusion, strip attention, and adaptive spatial fusion to strengthen weak targets[4–8]. These approaches are important comparators for PIR-SBFR because they test whether better connectivity or generic learned weighting is sufficient.

Most feature-pyramid weighting mechanisms are estimated from visual activations and optimized only through the detection objective. This design can adapt to scene content, but it does not assign a specific meaning to a routing weight. A larger P3 weight may indicate that the scene contains more small objects, that fine activations happen to be stronger, or that the model has learned a dataset-specific correlation. More importantly, feature content alone does not distinguish nominal resolution from usable resolution: P3 retains a fine spatial grid after blur or resampling, even when much of its high-frequency evidence has been removed. Generic attention and adaptive fusion address *where* or *how much* to fuse without explicitly modeling *whether* a level remains physically reliable under the acquisition condition.

2.3. Acquisition-Aware and Robust Detection

Acquisition-aware detection provides a second relevant research line. GSDDet demonstrates that spatial-resolution information can support remote-sensing object detection[14], while Meta-YOLO uses platform telemetry to adapt spatial sampling in a real-time detector[15]. FiLM offers a more general feature-wise affine conditioning mechanism[16]. More broadly, WILDS shows that naturally occurring distribution shifts, including shifts in satellite imagery, can substantially reduce out-of-distribution accuracy[17]. Together, these works motivate the use of information beyond RGB appearance when the deployed observation process differs from the training distribution.

However, supplying metadata is not equivalent to defining feature reliability. Direct concatenation, feature-wise modulation, or a general conditioning network may exploit metadata effectively, but the learned response is not necessarily monotonic with sampling quality, optical transfer, or noise. Such encoders can also learn correlations between metadata and scene category that do not reflect the intended image-formation relationship. Missing fields create a further problem: substituting zeros or a fixed value without an availability indicator may be interpreted as a valid acquisition state. Three requirements remain insufficiently addressed in combination: an explicit level-wise reliability quantity, separation of this quantity from scene-scale demand, and a fallback for incomplete metadata. Robustness evidence should include correspondence-breaking controls on degraded images, such as shuffled metadata, in addition to clean-set accuracy.

2.4. Positioning of PIR-SBFR

PIR-SBFR draws on these research lines but assigns explicit variables to the router. The analytic branch constrains cross-level weighting to represent acquisition-dependent P3–P5 reliability. Its availability-aware monotonic prior also permits a direct test of whether sample–metadata correspondence matters. Scale supervision gives the visual path an object-scale target, so it estimates scene demand and supplies a bounded correction to the physical prior. The acquisition descriptors remain active at inference, which also distinguishes the method from degradation augmentation.

The method retains the DRFB backbone and FACH prediction head of YOLO-DSF because receptive-field preservation and task coupling address problems complementary to routing. It replaces the original visual-only SBFR neck, adds an unweighted P5 bypass to protect coarse structural evidence, and incorporates missing-metadata and degradation-consistency mechanisms. The router is formulated around two inspectable quantities: scale demand and acquisition-dependent reliability. The experiments compare this formulation with generic fusion, direct metadata conditioning, GSD-only attention, and degradation augmentation. Negative controls and unseen degradations test whether the measured behavior agrees with the proposed mechanism.

3. Materials and Methods

3.1. Problem Formulation and Overall Framework

The reliability of a feature-pyramid level is not determined by image content alone. For the same ground object, a coarser ground sampling distance (GSD) reduces its pixel footprint, a lower modulation transfer function (MTF) or a wider point-spread function (PSF) attenuates high-frequency structure, and a lower signal-to-noise ratio (SNR) reduces the stability of local responses. These effects are strongest in the fine-resolution branch and can make a visually learned router unreliable outside the acquisition conditions represented during training. We formulate multiscale fusion as acquisition-conditioned reliability estimation with a constrained analytic prior.

The proposed detector retains the Dilated Receptive Field Block (DRFB) backbone and Feature-Aware Coupled Head (FACH) of YOLO-DSF, while replacing its visual-only SBFR neck with Physical Imaging Reliability-Guided Scale-Biased Feature Reweighting (PIR-SBFR). For an input image I, an acquisition descriptor $\mathbf{m}$, and its availability mask $\mathbf{a}$, the forward pass is

$$\{P_3, P_4, P_5\} = \mathcal{B}_{\mathrm{DRFB}}(I), \qquad \{F_3, F_4, F_5\} = \mathcal{N}_{\mathrm{PIR}}(\{P_i\}, \mathbf{m}, \mathbf{a}), \tag{1}$$

$$\widehat{Y} = \mathcal{H}_{\mathrm{FACH}}(F_3, F_4, F_5), \tag{2}$$

where P_3, P_4, and P_5 have strides 8, 16, and 32, respectively. As summarized in Figure 2, PIR-SBFR combines two complementary signals. An analytic branch converts the available acquisition variables into a monotonic reliability prior, whereas a visual branch estimates an image-dependent residual and the scene scale distribution. Their normalized sum controls the contribution of each pyramid level. A direct P5 bypass is retained so that physical reweighting does not remove the global structural evidence required by airports, bridges, dams, and other large continuous objects.

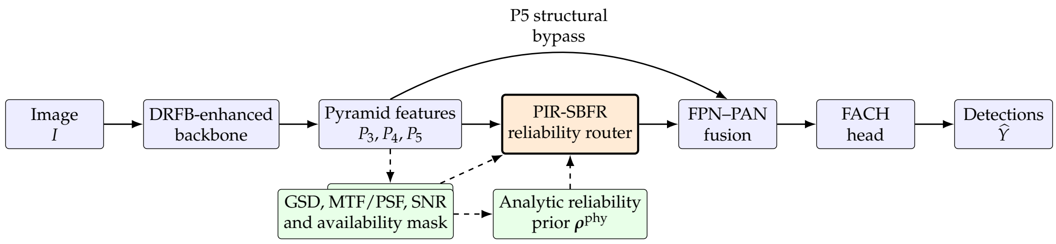

Figure 2. Overall framework of the proposed detector. PIR-SBFR combines an analytic acquisition-condition prior with a scale-supervised visual residual to route P3–P5 features. Solid arrows denote the detection path, and dashed arrows denote conditioning signals.

3.2. Dilated Receptive Field Backbone

DRFB is retained as the feature extractor because reliability-aware routing cannot recover spatial evidence that has already been removed by early downsampling. For an input tensor $X^{(l-1)} \in \mathbb{R}^{H\times W\times C}$, a 1×1 convolution first reduces the channel dimension:

$$X^{(l)}_{\mathrm{red}} = \sigma\Big(\mathrm{BN}\Big(\mathrm{Conv}^{C\to C/2}_{1\times 1}(X^{(l-1)})\Big)\Big), \tag{3}$$

where σ denotes the activation function. Two 3×3 dilated convolutions then model local and wider context without changing the feature resolution:

$$D^{(l)}_r = \sigma\Big(\mathrm{BN}\Big(\mathrm{DConv}^{r}_{3\times 3}(X^{(l)}_{\mathrm{red}})\Big)\Big), \quad r \in \{2, 3\}. \tag{4}$$

Their concatenated response is compressed, restored to C channels, and added to the identity path:

$$X^{(l)}_{\mathrm{ctx}} = \mathrm{Conv}_{\mathrm{cat}}\Big([D^{(l)}_2, D^{(l)}_3]\Big), \tag{5}$$

$$X^{(l)} = X^{(l-1)} + \phi^{C/2\to C}_{1\times 1}\Big(X^{(l)}_{\mathrm{ctx}}\Big). \tag{6}$$

The identity branch preserves fine boundary evidence, while the two dilation rates supply context for targets whose appearance is weak or fragmented. Dilated convolution expands the receptive field without reducing spatial resolution, and the residual path stabilizes the preservation of identity information[18,19]. The module is placed before severe downsampling so that the subsequent PIR-SBFR router receives spatially meaningful P3 and P4 features.

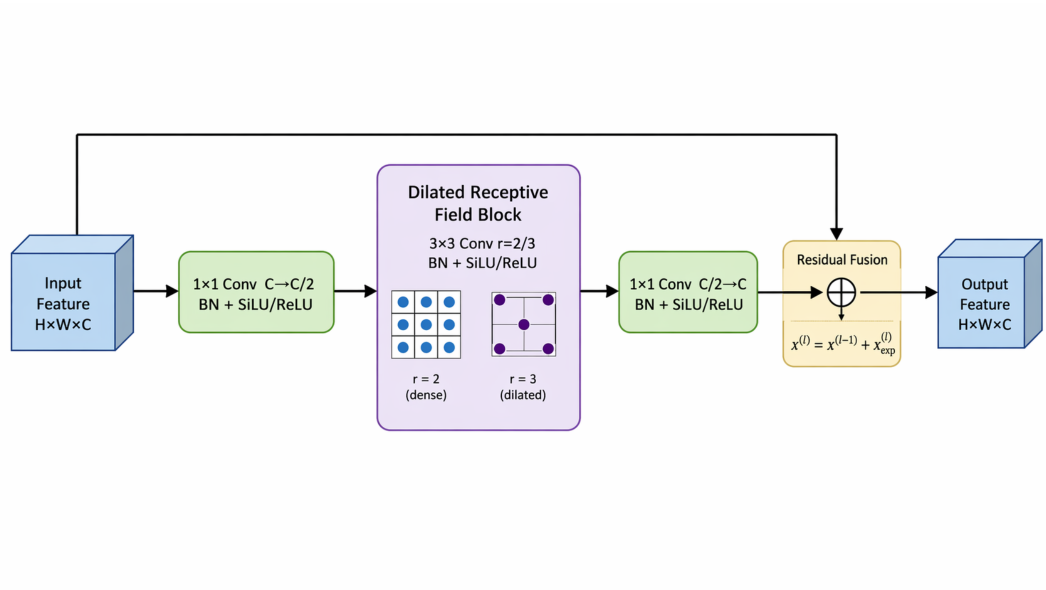

Figure 3. DRFB structure with channel compression, dual-rate dilated context modeling, channel restoration, and residual preservation.

3.3. *Physical Imaging Reliability-Guided Scale-Biased Feature Reweighting*

Figure 4 expands the PIR-SBFR reliability router in Figure 2. The visual path aligns and pools P3–P5 features to estimate a scale-supervised residual, while the analytic path converts available acquisition descriptors into level-wise physical reliability. Their logit-space fusion produces normalized routing weights for feature reweighting before FPN–PAN fusion. The separate P5 projection preserves unweighted coarse structural evidence.

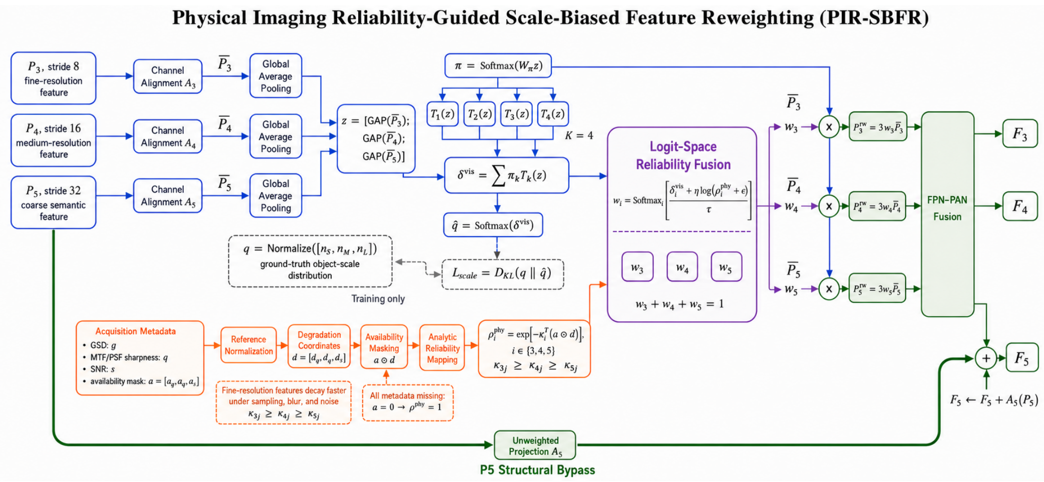

Figure 4. Detailed architecture of PIR-SBFR. The blue path estimates the visual residual and training-only scale-distribution objective. The orange path derives the analytic physical reliability prior from GSD, MTF/PSF, SNR, and availability masks. The purple block produces normalized P3–P5 routing weights. The green path performs feature reweighting, FPN–PAN fusion, and the P5 structural bypass. Dashed connections denote training-only supervision or explanatory constraints.

3.3.1. Acquisition Descriptor and Analytic Reliability Prior

For each image, the acquisition descriptor is written as

$$\mathbf{m} = [g, q, s]^{\mathsf{T}}, \qquad \mathbf{a} = [a_g, a_q, a_s]^{\mathsf{T}} \in \{0, 1\}^3, \tag{7}$$

where g is GSD (or relative GSD when only a reference scale is available), q is the MTF value at the Nyquist frequency or an equivalent PSF-derived sharpness measure, and s is

SNR in decibels. The binary mask a_j records whether the corresponding variable is available. In the auxiliary flight experiment, absolute camera calibration is not required: relative GSD is computed from the attitude-corrected slant range as $g = (H/\cos\theta)/(H_{\text{ref}}/\cos\theta_{\text{ref}})$, where H is recorded height and θ is the off-nadir angle derived from the synchronized attitude/IMU fields. When MTF/PSF or SNR is not recorded, its mask is set to zero; the missing quantity is not estimated from the image labels.

The three variables have different units and directions. We convert them to non-negative degradation coordinates relative to the clean training reference $g_{\text{ref}} = 1$, $q_{\text{ref}} = 0.50$, and $s_{\text{ref}} = 30$ dB:

$$d_g = \max\left(0, \log\frac{g}{g_{\text{ref}}}\right), \quad d_q = \max\left(0, 1 - \frac{q}{q_{\text{ref}}}\right), \quad d_s = \max\left(0, \frac{s_{\text{ref}} - s}{s_{\text{ref}}}\right). \tag{8}$$

Thus, coarser sampling, lower optical transfer, and lower SNR increase the corresponding degradation coordinate. Let $\mathbf{d} = [d_g, d_q, d_s]^\top$. The physical reliability assigned to pyramid level $i \in \{3, 4, 5\}$ is

$$\rho_i^{\text{phy}} = \exp\left[-\boldsymbol{\kappa}_i^\top(\mathbf{a} \odot \mathbf{d})\right], \tag{9}$$

where $\boldsymbol{\kappa}_i \geq 0$ is fixed by the pyramid stride $r_i \in \{8, 16, 32\}$ and is not tuned on the test set:

$$\boldsymbol{\kappa}_i = \frac{8}{r_i}[1, 1, 1]^\top, \qquad \begin{bmatrix} \boldsymbol{\kappa}_3^\top \\ \boldsymbol{\kappa}_4^\top \\ \boldsymbol{\kappa}_5^\top \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0.5 & 0.5 & 0.5 \\ 0.25 & 0.25 & 0.25 \end{bmatrix}. \tag{10}$$

This parameter-free ordering expresses a simple imaging fact: fine-resolution responses lose reliability faster than contextual responses as sampling, blur, or noise deteriorates. The degradation coordinates already normalize the three physical variables, so equal within-level coefficients avoid introducing an additional metadata-specific fitting stage. The mapping is analytic and does not contain a trainable metadata encoder. Each term can be disabled independently, which permits separate GSD, MTF/PSF, and SNR ablations. If all three variables are missing, $\mathbf{a} = \mathbf{0}$ and $\boldsymbol{\rho}^{\text{phy}} = \mathbf{1}$, giving a neutral prior.

3.3.2. Scale-Supervised Visual Residual

The analytic prior cannot describe every scene-dependent factor, such as object density, land-cover clutter, or calibration error. PIR-SBFR learns a bounded visual correction from the pyramid itself. The three levels are first aligned in channel dimension and globally pooled:

$$\bar{P}_i = \mathcal{A}_i(P_i), \qquad \mathbf{z} = [\text{GAP}(\bar{P}_3); \text{GAP}(\bar{P}_4); \text{GAP}(\bar{P}_5)]. \tag{11}$$

A lightweight mixture with K branches produces the residual routing logits:

$$\boldsymbol{\pi} = \text{softmax}(W_\pi \mathbf{z}), \qquad \delta^{\text{vis}} = \sum_{k=1}^{K} \pi_k T_k(\mathbf{z}), \tag{12}$$

where T_k denotes the k-th residual branch. The associated visual scale estimate is

$$\widehat{\mathbf{q}} = \text{softmax}(\delta^{\text{vis}}). \tag{13}$$

For an image containing n_S, n_M, and n_L annotated instances, the target scale distribution is

$$\mathbf{q} = \frac{[n_S, n_M, n_L]^\mathsf{T} + \epsilon}{n_S + n_M + n_L + 3\epsilon}, \tag{14}$$

where the small, medium, and large groups follow the scale definition in the experimental protocol. The three components are associated with the fine, intermediate, and coarse pyramid levels, respectively: $q_S \leftrightarrow P_3$, $q_M \leftrightarrow P_4$, and $q_L \leftrightarrow P_5$. This soft scene-level correspondence does not assign each object to a single feature level; all three levels remain active in every forward pass. The scale-supervision term is

$$\mathcal{L}_{\text{scale}} = D_{\text{KL}}(\mathbf{q} \,\|\, \widehat{\mathbf{q}}). \tag{15}$$

The final detection loss constrains the predicted classes and boxes but does not uniquely identify the internal cross-level weights: different P3–P5 mixtures can produce similar output losses, permitting a visual router to converge to dataset-average weights or scene shortcuts. The auxiliary target reduces this ambiguity by tying the routing logits to an observable property of the training image. The target $\mathbf{q}$ is computed from annotations only during training. At inference, no object annotation or scale label is used; $\widehat{\mathbf{q}}$ and δ^{vis} are obtained entirely from the pyramid features.

Scale demand and feature reliability are deliberately kept separate. The scale target describes which feature resolutions are required by the objects present in the scene, whereas $\boldsymbol{\rho}^{\text{phy}}$ describes whether those resolutions remain trustworthy under the observed acquisition condition. Scale supervision alone cannot identify sampling, optical-transfer, or noise degradation, and the physical prior alone cannot infer whether a scene is dominated by small or large objects. Their fusion combines these complementary sources of cross-scale uncertainty.

3.3.3. Reliability Fusion, Feature Routing, and P5 Bypass

The analytic reliability and visual residual are fused in logit space. The final routing weight for pyramid level i is

$$w_i = \frac{\exp\Big[\big(\delta_i^{\text{vis}} + \eta \log(\rho_i^{\text{phy}} + \epsilon)\big)/\tau\Big]}{\sum_{j=3}^{5} \exp\Big[\big(\delta_j^{\text{vis}} + \eta \log(\rho_j^{\text{phy}} + \epsilon)\big)/\tau\Big]}, \quad \sum_{i=3}^{5} w_i = 1, \tag{16}$$

where η controls the physical-prior strength and τ is the routing temperature. The residual term can correct an imperfect prior, but the analytic term preserves the expected monotonic response to acquisition degradation. Unlike direct concatenation or feature-wise linear modulation, this construction exposes metadata influence through an explicit reliability quantity. It does not use an unrestricted metadata embedding.

This dynamic rule resolves cases for which fixed fusion is intrinsically ambiguous. In a clear, finely sampled scene containing many small objects, the visual demand favors P3 and the physical prior leaves its reliability high. For the same scale demand under coarser sampling, stronger blur, or lower SNR, ρ_3^{phy} decreases more rapidly and part of the routing mass moves toward P4 and P5. Conversely, a large-structure-dominant scene can favor P5 through the visual residual even when the acquisition quality is high. The router thus changes the relative contribution of all three levels per image instead of permanently selecting or suppressing any one level.

To keep the average feature magnitude unchanged when the three weights are equal, the aligned features are reweighted as

$$P_i^{\text{rw}} = 3 w_i \bar{P}_i, \qquad i \in \{3, 4, 5\}. \tag{17}$$

The reweighted levels are then processed by the original FPN–PAN operations:

$$\{F_3, F_4, F_5\} = \mathcal{N}_{\mathrm{FPN-PAN}}(P_3^{\mathrm{rw}}, P_4^{\mathrm{rw}}, P_5^{\mathrm{rw}}). \tag{18}$$

Because large continuous structures depend strongly on coarse global context, an unweighted projection of P_5 is added to the coarse output:

$$F_5 \leftarrow F_5 + \mathcal{A}_5(P_5). \tag{19}$$

This bypass prevents the router from trading large-structure evidence for small-object sensitivity and makes the claimed scope of the method testable through AP_L and category-level analysis.

3.3.4. Missing-Metadata Robustness and Degradation Consistency

Metadata are not assumed to be complete at deployment. During training, each acquisition variable and its mask are dropped with probability p_{drop}. The analytic branch uses only $\mathbf{a} \odot \mathbf{d}$, so a missing variable contributes neither an arbitrary zero measurement nor a learned test-set imputation. With complete metadata loss, $\boldsymbol{\rho}^{\mathrm{phy}} = \mathbf{1}$, and the routing equation reduces to the scale-supervised visual router. This same fallback is used at inference, without retraining.

Training additionally uses paired clean and degraded views. The training degradations comprise isotropic Gaussian blur and Poisson noise; the unseen PSF and noise families used for robustness evaluation are excluded from model selection. Let $(\mathbf{p}_r, \mathbf{b}_r)$ and $(\mathbf{p}'_r, \mathbf{b}'_r)$ denote the class-probability vector and normalized box for the same positive assignment r in the clean and degraded views. The degradation-consistency loss is

$$\mathcal{L}_{\mathrm{cons}} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \left[D_{\mathrm{SKL}}(\mathbf{p}_r, \mathbf{p}'_r) + \|\mathbf{b}_r - \mathbf{b}'_r\|_1 \right], \tag{20}$$

where D_{SKL} is the symmetric Kullback–Leibler divergence and $\mathcal{R}$ is the set of matched positive assignments. Consistency is imposed on predictions, not on routing weights, because the router is expected to change when feature reliability changes. The complete objective is

$$\mathcal{L} = \frac{1}{2} \left(\mathcal{L}_{\mathrm{det}}(I) + \mathcal{L}_{\mathrm{det}}(I') \right) + \lambda_{\mathrm{scale}} \mathcal{L}_{\mathrm{scale}} + \lambda_{\mathrm{cons}} \mathcal{L}_{\mathrm{cons}}. \tag{21}$$

The reported experiments use $\eta = 1$, $\tau = 1$, $K = 4$, $\lambda_{\mathrm{scale}} = 0.1$, $\lambda_{\mathrm{cons}} = 0.1$, and $p_{\mathrm{drop}} = 0.25$ as fixed implementation settings. No numerical sensitivity claim is made because separable validation traces were not retained.

3.4. Feature-Aware Coupled Head

The FACH prediction head is retained after PIR-SBFR because small remote-sensing objects require semantic evidence for classification and fine spatial evidence for localization. For neck output $F_i \in \mathbb{R}^{H_i \times W_i \times C}$, global average pooling produces $\mathbf{z}_i = \mathrm{GAP}(F_i)$. Channel reweighting gives

$$F_i^c = F_i \odot \sigma(W_c \mathbf{z}_i + \mathbf{b}_c). \tag{22}$$

Routing coefficients $\boldsymbol{\phi}_i = \mathrm{softmax}(W_r \mathbf{z}_i + \mathbf{b}_r)$ mix K_h feature transformations while preserving the identity path:

$$F_i^{\mathrm{out}} = F_i + \sum_{j=1}^{K_h} \phi_{ij} T_j(F_i^c). \tag{23}$$

A coupled convolution forms a shared task representation before the final task separation:

$$G_i = \mathcal{C}_{\text{coupled}}(F_i^{\text{out}}), \qquad y_i^{\text{cls}} = H_{\text{cls}}(G_i), \quad y_i^{\text{reg}} = H_{\text{reg}}(G_i). \tag{24}$$

The identity connection limits the risk that dynamic coupling removes weak localization cues, while the shared representation permits classification and regression information to interact before the final prediction layers. This design is related to input-dependent kernel aggregation, attention-based dynamic heads, and task-aligned classification–localization coupling[20–22]. Importantly, FACH operates after the acquisition-conditioned routing; it does not consume metadata and remains unchanged between correct-, noisy-, and missing-metadata evaluations.

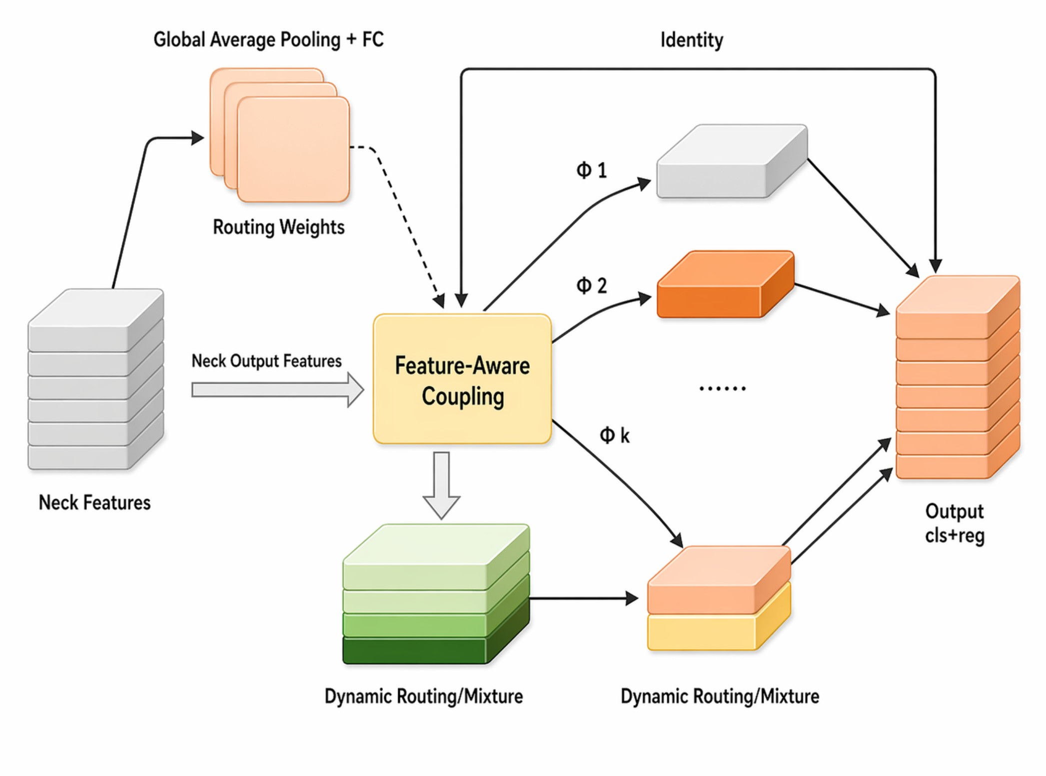

Figure 5. FACH structure with global routing, identity preservation, dynamic feature coupling, and final classification/regression separation.

3.5. Experimental Design and Evaluation Protocol

3.5.1. Datasets, Acquisition Conditions, and Degradation Generation

Experiments are organized into four complementary settings. DIOR is the principal clean-set benchmark. It contains 23,463 optical remote-sensing images, 192,472 annotated instances, and 20 object categories[2]. The distributed `train.txt`, `val.txt`, and `test.txt` files contain 5,862, 5,863, and 11,738 image identifiers, respectively. All models use these fixed identifiers.

Bounding boxes are measured after resizing to 640×640. Small objects have area below 32^2 pixels. Medium objects have area from 32^2 to below 96^2 pixels, and large objects have area of at least 96^2 pixels. These definitions govern n_S, n_M, n_L, $\text{AP}_{S/M/L}$, and P3–P5 scale supervision.

AI-TOD-v2 provides an independent tiny-object benchmark with 28,036 images, 752,754 instances, and eight categories[23]. Its mean absolute object size is 12.7 pixels. The `aitodv2_train.json`, `aitodv2_val.json`, and `aitodv2_test.json` records contain

11,214, 5,607, and 11,215 images, respectively. All methods use the same annotation release and identifiers.

AI-TOD-v2 uses four reporting intervals: very tiny (2–8 pixels), tiny (8–16 pixels), small (16–32 pixels), and medium (32–64 pixels). Its routing target is $\mathbf{q}_{\text{AITOD}} = \text{Normalize}([n_{VT} + n_T, n_S, n_M])$. Very-tiny and tiny objects supervise P3, small objects supervise P4, and medium objects supervise P5. The four original intervals remain unchanged for evaluation.

Table 1. Public-dataset image partitions used by every compared method. Counts refer to unique image identifiers in the listed distribution files.

Dataset	Partition	Images	Distribution record
DIOR	Training	5,862	`train.txt`
DIOR	Validation	5,863	`val.txt`
DIOR	Test	11,738	`test.txt`
AI-TOD-v2	Training	11,214	`aitodv2_train.json`
AI-TOD-v2	Validation	5,607	`aitodv2_val.json`
AI-TOD-v2	Test	11,215	`aitodv2_test.json`

Public benchmark images do not provide uniform per-image sensor telemetry. Controlled acquisition experiments therefore use the known parameters of each applied image-formation transformation. Let $x \in [0,1]^{640\times640\times3}$ denote the resized image. Relative GSD r is implemented by antialiased bicubic downsampling to $\text{round}(640/r)$ pixels per side. Bicubic reconstruction then restores the 640×640 input size.

Optical transfer q is implemented by an isotropic Gaussian PSF. Its standard deviation follows $\sigma_q = \sqrt{-2\log q}/\pi$, with kernel width $2\lceil 3\sigma_q \rceil + 1$. Shot noise is applied last. For target SNR s, the Poisson rate is $\lambda = \overline{x}\,10^{s/10}/\overline{x^2}$. The degraded sample is $\text{Poisson}(\lambda x)/\lambda$, clipped to $[0,1]$. The fixed order is sampling, blur, and shot noise. Random seeds are paired across models.

The clean reference uses relative GSD $= 1$, MTF $= 0.50$, and SNR $= 30$ dB. Each training source supplies one clean view and one degraded view. Thus, the clean-to-degraded pair ratio is 1:1. A batch contains 16 source images. The two views are evaluated separately, and their losses are averaged before one optimizer update.

The degraded view uses Gaussian blur only with probability 0.35, Poisson noise only with probability 0.35, and both operators with probability 0.30. For blur, $q \sim U(0.15, 0.45)$. For noise, $s \sim U(10, 28)$ dB. Relative GSD remains one during training. Each metadata field is independently masked with probability $p_{\text{drop}} = 0.25$.

Spatial and photometric augmentation parameters are sampled once before the clean and degraded views are formed. Both views therefore share the same crop, scale, flip, and labels. The degradation random state is determined by the training seed, epoch, and image identifier. Motion blur, disk PSFs, anisotropic blur, speckle noise, stripe noise, and resolution changes are excluded from training.

Table 2. Sampling of the paired training views. A dash indicates that the corresponding degradation operator is not applied.

View or mode	Probability	Relative GSD	MTF parameter	SNR parameter
Clean view	1.00	1	0.50 reference	30 dB reference
Blur-only degraded view	0.35	1	$U(0.15, 0.45)$	–
Noise-only degraded view	0.35	1	–	$U(10, 28)$ dB
Joint degraded view	0.30	1	$U(0.15, 0.45)$	$U(10, 28)$ dB

The factorial robustness grid contains all $3 \times 3 \times 3 = 27$ combinations of GSD $\{1, 2, 3\}$, MTF $\{0.50, 0.30, 0.15\}$, and SNR $\{30, 20, 10\}$ dB. It is evaluated for YOLOv11n, YOLO-DSF, and PIR-SBFR Full. This design yields 81 model–condition observations. Each value is a three-seed mean.

Metadata correspondence is assessed on a separate mixed-degradation DIOR control set. Its images use non-reference GSD, MTF, or SNR settings from the controlled space. Image content and model checkpoints remain fixed across metadata conditions. Only descriptor values and availability masks change. Consequently, $\mathbf{a} \odot \mathbf{d}$ remains informative. Only within-set changes relative to correctly paired metadata are reported.

The OOD test is separated from the factorial grid. Nine held-out conditions are excluded from training, early stopping, hyperparameter selection, and model selection. Three use unseen PSFs: disk defocus with radius 3 pixels, 9-pixel motion blur with random orientation, and anisotropic Gaussian blur with $\sigma_x = 2.5$, $\sigma_y = 0.6$, and random orientation.

Two held-out conditions use unseen noise. They comprise multiplicative speckle with $\sigma = 0.12$, and stripe noise of amplitude 0.08 with read noise of $\sigma = 0.02$. Three conditions use antialiased 1.5×, 2.5×, and 4× sampling changes. The final condition combines 2.5× sampling, an unseen PSF, and unseen noise.

The aggregate record identifies the joint operator families but not its sampled PSF orientation. This condition is therefore a composite stress test and does not support pixel-exact reconstruction. All other held-out transformations are fully specified. Every OOD value is a three-seed mean.

The auxiliary flight collection contains eight complete videos recorded from 5.2 to 30.0 m. A DJI Mavic 3 Enterprise wide camera was used. It has a 4/3 CMOS sensor and a 24 mm equivalent lens. Video was recorded at 3840 × 2160, 30 frames/s, using H.264 encoding. Height, attitude, and IMU data were logged at 10 Hz and synchronized by timestamp.

Every sixth video frame was annotated, giving a 5 frames/s annotation rate. Horizontal boxes were created in CVAT for vehicle, person, and ship. Tracking proposals were permitted, but every box was manually checked. A second annotator independently reviewed 10% of frames. Class conflicts, omissions, and boundary disagreements were adjudicated.

Recorded height and attitude are converted to relative GSD using Section 3.3.1. Unavailable MTF/PSF and SNR fields are masked rather than inferred. Complete videos are assigned before frame extraction, preventing adjacent-frame leakage. Table 3 reports the collection inventory.

Table 3. Inventory of the auxiliary flight-video collection. Raw and annotated frame counts refer to 30 and 5 frames/s, respectively.

Video	Height (m)	Scene	Duration (s)	Raw frames	Annotated frames	Instances	Classes
F01	5.2–8.4	Parking area	120	3,600	600	4,238	vehicle, person
F02	7.8–10.0	Campus road	108	3,240	540	3,691	vehicle, person
F03	10.1–14.7	Road intersection	132	3,960	660	4,884	vehicle, person
F04	11.8–15.0	Industrial road	126	3,780	630	4,516	vehicle, person
F05	15.2–19.8	Road and parking area	144	4,320	720	5,307	vehicle, person
F06	20.1–24.8	Riverside road	138	4,140	690	4,762	vehicle, person, ship
F07	25.0–27.8	Industrial parking area	156	4,680	780	5,941	vehicle, person
F08	27.9–30.0	Riverside industrial area	150	4,500	750	5,608	vehicle, person, ship
Total	5.2–30.0	Eight complete videos	1,074	32,220	5,370	38,947	three classes

Table 4 defines every flight-video partition. Buffer videos are excluded from both fitting and evaluation. In eight-fold leave-one-video-out evaluation, each video is tested once while the other seven form the training set. Every flight experiment uses the fixed 200-epoch schedule without validation-based model selection.

Table 4. Complete-video partitions for the auxiliary flight experiments.

Protocol	Training videos	Test videos	Excluded buffer videos
Same altitude, unseen flight	F01–F03, F05–F08	F04	None
Unseen high altitude	F01–F05	F07–F08	F06
Unseen low altitude	F05–F08	F01–F02	F03–F04
Eight-fold leave-one-video-out	Remaining seven videos in each fold	F01–F08, one per fold	None

The collection supports only paired within-collection checks and is not presented as a public benchmark. Its results remain separate from public-dataset AP ranges.

3.5.2. Implementation and Training Details

The models are implemented in PyTorch 2.8.0 and trained on Ubuntu 22.04 with CUDA 12.8. Training uses one NVIDIA RTX 4090D GPU with 24 GB memory and an AMD EPYC 9754 processor. All competing modules use the same YOLOv11n codebase and are trained from scratch. Images are resized to 640×640, and training lasts 200 epochs.

Stochastic gradient descent uses an initial learning rate of 0.005, momentum 0.937, weight decay 5×10^{-4}, and three warmup epochs. Mosaic is applied with probability 1.0 for epochs 1–180 and disabled for the final 20 epochs. HSV perturbation is applied to every training sample. Hue offset is drawn from $U(-0.015, 0.015)$. Saturation and value multipliers are drawn from $U(0.30, 1.70)$ and $U(0.60, 1.40)$, respectively.

The affine pipeline uses a scale factor drawn from $U(0.50, 1.50)$ and translation up to 0.10 of image width or height. Horizontal flipping has probability 0.50. Vertical flipping, rotation, shear, perspective transformation, MixUp, and Copy-Paste are disabled. Three paired seeds, 2023, 2024, and 2025, are used for principal models and physical-component ablations.

The reported PIR-SBFR configuration uses $\eta = 1$, $\lambda_{\text{scale}} = 0.1$, $\lambda_{\text{cons}} = 0.1$, $\tau = 1$, $K = 4$, and $p_{\text{drop}} = 0.25$. Separable validation traces were not retained, so no numerical hyperparameter-sensitivity analysis is reported.

Efficiency is measured separately on an RTX 4090 with FP16, batch size 1, and 640×640 inputs. The input tensor is preallocated on the GPU. Each model receives 200 warm-up passes, followed by five runs of 1,000 timed passes. CUDA is synchronized before and after every timed run. We report the median of the five run-wise means, and FPS is 1000/latency(ms).

The measurement includes network forward propagation only. Disk access, image decoding, resizing, host-to-device transfer, NMS, and output serialization are excluded. Peak VRAM is measured after resetting PyTorch peak-memory statistics. These values are consequently forward-latency measurements rather than end-to-end deployment latency.

3.5.3. Metrics and Statistical Analysis

Unless stated otherwise, AP denotes COCO-style mAP averaged over IoU thresholds from 0.50 to 0.95. We additionally report AP50, AP75, scale-specific AP ($\text{AP}_{S/M/L}$), and scale-specific average recall ($\text{AR}_{S/M/L}$). Precision, Recall, and F1 are included for continuity with the original YOLO-DSF evaluation. AI-TOD-v2 additionally uses AP_{VT}, AP_{T}, AP_{S}, and AP_{M} according to its pixel-size intervals.

Two uncertainty sources are reported separately. A paired t-interval over the three common seeds estimates training-run variation. A paired bootstrap estimates test-sample uncertainty for a fixed model contrast. Predictions are stored as COCO-format records containing image identifier, category, box, confidence, and model seed.

For each bootstrap replicate, test image identifiers are sampled with replacement to the original test-set size. The same sampled identifiers are used for both models. Repeated identifiers are copied and remapped before COCO evaluation, preserving their multiplicity. COCO AP is recomputed for every replicate rather than averaged from per-image scores.

We use 10,000 replicates, random seed 20260718, and percentile intervals at 2.5% and 97.5%. The flight analysis resamples complete videos rather than frames. The paired procedure is implemented in `scripts/bootstrap_paired_coco.py`. Absolute changes are expressed in percentage points, not relative percentages.

4. Results

4.1. Clean-Set Performance and Cross-Dataset Generalization

4.1.1. DIOR Main Results

Table 5 reports the three-seed DIOR results. PIR-SBFR Full reaches 65.52 ± 0.42 AP, compared with 62.98 ± 0.43 for YOLO-DSF and 57.31 ± 0.47 for YOLOv11n. Relative to YOLO-DSF, the full model gains 2.54 pp AP, 3.08 pp AP75, 3.53 pp AP_S, 2.63 pp AP_M, and 4.18 pp AR_S. The 1.62 pp AP_L increase is smaller than the small-object gain but is important because the original YOLO-DSF reduces large-object AP relative to YOLOv11n. The direct P5 path improves this large-structure weakness without reversing the small-object benefit.

The joint scale-supervision/P5-bypass configuration raises AP from 62.98 to 63.69, whereas adding the physical prior, visual correction, metadata dropout, and consistency loss raises it to 65.52. The improvement is consequently not explained by the initial scale-routing modification alone. Figure 1(a) summarizes the cross-dataset gains, and Figure 1(b) places them against computational cost.

Table 5. Three-seed results on DIOR. All entries are percentages; AP is shown as mean $\pm$ sample SD. Best values are bold.

Model	P	R	F1	AP50	AP75	AP	AP_S	AP_M	AP_L	AR_S
YOLOv11n	86.67	71.93	78.62	77.82	61.24	57.31 ± 0.47	38.64	60.52	72.04	45.23
YOLO-DSF	90.12	80.77	85.19	82.31	67.63	62.98 ± 0.43	44.08	65.79	70.57	52.03
DSF + scale supervision + P5 bypass	89.93	81.29	85.39	83.02	68.41	63.69 ± 0.43	45.21	66.32	70.91	53.12
PIR-SBFR, no consistency	90.51	82.31	86.21	84.51	70.32	65.17 ± 0.42	47.18	68.01	71.98	55.72
PIR-SBFR Full	**90.72**	**82.61**	**86.47**	**84.92**	**70.71**	$\mathbf{65.52 \pm 0.42}$	**47.61**	**68.42**	**72.19**	**56.21**

4.1.2. AI-TOD-v2 Tiny-Object Results

Table 6 confirms that the gain transfers to a dataset dominated by much smaller targets. PIR-SBFR Full improves AP from 29.61 to 32.13 over YOLO-DSF. The increases are 2.60 pp for very-tiny objects, 2.63 pp for tiny objects, 2.61 pp for small objects, and 3.49 pp for average recall. This consistent scale-wise pattern is stronger evidence for the intended use case than AP50 alone. The remaining medium-object gain is modest (1.19 pp), which is consistent with the router concentrating most of its additional capacity on low-pixel targets.

Table 6. Three-seed results on AI-TOD-v2. All entries are percentages; AP is shown as mean $\pm$ sample SD.

Model	AP50	AP75	AP	AP_{VT}	AP_T	AP_S	AP_M	AR	AR_{VT}
YOLOv11n	48.23	24.06	25.82 ± 0.37	10.84	24.13	36.52	45.31	36.42	18.84
YOLO-DSF	53.72	28.34	29.61 ± 0.37	14.62	28.41	39.71	45.02	41.13	23.92
PIR-SBFR Full	**56.92**	**31.21**	$\mathbf{32.13 \pm 0.36}$	**17.22**	**31.04**	**42.32**	**46.21**	**44.62**	**27.43**

4.2. Comparison with Alternative Fusion and Conditioning Strategies

To test whether the gain can be explained by a stronger generic neck or arbitrary metadata injection, Table 7 compares all methods under the same backbone, input size, augmentation, epoch count, and evaluation code. The controls include BiFPN[11], ASFF[12], AFPN[13], a GSD-guided attention implementation following the scale-conditioning principle of GSDDet[14], direct metadata concatenation, and feature-wise linear modulation[16]. We also include an implementation inspired by metadata-guided aerial detection[15].

PIR-SBFR Full exceeds BiFPN, ASFF, and AFPN by 3.65, 3.24, and 2.95 pp AP, respectively. It also outperforms direct concatenation by 1.54 pp, GSD-only attention by 1.30 pp, Metadata-FiLM by 1.12 pp, and the analytic GSD+MTF+SNR prior without the complete visual/consistency design by 0.65 pp. These comparisons isolate the contribution of the proposed analytic-prior/visual-residual combination: metadata are useful, but an arbitrary conditional encoder does not reproduce the full result.

Table 7. Same-protocol comparison on DIOR. All accuracy values are percentages.

Method	AP50	AP75	AP	AP_S	Params (M)	GFLOPs
YOLOv11n	77.82	61.24	57.31	38.64	2.584	6.47
PAN-FPN + DRFB/FACH	80.77	65.51	61.18	42.14	3.462	8.08
BiFPN[11]	81.41	66.32	61.87	42.87	3.707	8.57
ASFF[12]	81.83	66.81	62.28	43.72	3.779	8.73
AFPN[13]	82.01	67.08	62.57	43.96	3.824	8.86
YOLO-DSF	82.31	67.63	62.98	44.08	3.628	8.43
Physics-degradation augmentation	82.82	68.04	63.37	44.65	3.628	8.43
Metadata direct concatenation	83.36	68.62	63.98	45.62	3.843	8.78
GSD-conditioned scale attention[14]	83.62	68.91	64.22	46.08	3.752	8.61
Metadata-FiLM[15,16]	83.81	69.22	64.40	46.21	3.878	8.81
Analytic GSD+MTF+SNR fusion	84.11	69.88	64.87	46.77	3.821	8.69
PIR-SBFR, no consistency	84.51	70.32	65.17	47.18	3.897	8.76
PIR-SBFR Full	**84.92**	**70.71**	**65.52**	**47.61**	3.942	8.82

4.3. Comparison with General and Remote-Sensing Detectors

Table 8 places PIR-SBFR against the detector baselines retained from the original same-split DIOR evaluation. These models use the same 640×640 input resolution and dataset split; the comparison is reported at AP50 because this is the common metric available for every retained baseline. This table is supplementary to the stricter AP50:95 comparisons used throughout the principal experiments. PIR-SBFR Full reaches 84.92% AP50, 2.61 pp above YOLO-DSF and 3.12 pp above the strongest non-DSF detector in the table, YOLOv9t. Its Recall of 82.61% is also 2.21 pp above RSI-YOLO, the strongest competing Recall result. Differences in architecture and training implementation limit this table to a broad competitiveness check. The controlled neck and metadata comparisons in Table 7 provide the more direct attribution test.

Table 8. Comparison with general and remote-sensing object detectors on the common DIOR split. All values are percentages.

Model	Precision	Recall	AP50
YOLOv8n[24]	86.60	75.20	80.20
YOLOv9t[25]	88.80	75.60	81.80
YOLOv10n[26]	86.50	75.00	80.90
YOLOv11n[9]	86.67	71.93	77.82
YOLOv12n[27]	89.60	75.40	81.50
RetinaNet[28]	83.40	71.10	73.90
RT-DETR[29]	87.50	77.40	81.70
RSI-YOLO[4]	78.80	80.40	79.40
RepDarkNet-B3[30]	70.00	77.00	75.50
MSA-YOLO[5]	74.40	66.20	68.90
YOLO-DSF	90.12	80.77	82.31
PIR-SBFR Full	**90.72**	**82.61**	**84.92**

4.4. Component Attribution

Table 9 uses three independent runs for every ablation. Degradation augmentation provides 0.39 pp over YOLO-DSF, while the joint introduction of scale-distribution supervision and the P5 bypass (A2) provides 0.71 pp. A2 raises AP_S from 44.08 to 45.21 (+1.13 pp), compared with a 0.34 pp change in AP_L; the main-results table also shows a 1.09 pp increase in AR_S. This scale-concentrated pattern is consistent with the intended role of scale-constrained routing. Because A2 activates both scale supervision and the bypass, the 0.71 pp gain supports only their joint configuration; it is not an isolated causal estimate for $\mathcal{L}_{\text{scale}}$. Individually, GSD, MTF/PSF, and SNR reliability contribute 1.24, 1.13, and 0.95 pp, while their analytic combination reaches 64.87 AP. Compared with A2, the combined analytic prior adds 1.18 pp AP and 1.56 pp AP_S, showing that the initial scale-routing configuration does not capture acquisition-dependent reliability. The analytic combination is also 0.89 pp better than direct concatenation and 0.47 pp better than Metadata-FiLM, even before adding the visual residual. The visual residual then raises AP to 65.17, and degradation consistency adds a further 0.35 pp. The complete AP_S progression, from 44.08 to 47.61, shows that the primary effect occurs at the intended scale.

The ablation is not artificially monotonic: MTF-only and SNR-only variants are slightly below the GSD-only variant, and degradation augmentation yields a small gain. The conclusion rests on the combined analytic prior, its visual correction, and consistency regularization. It does not require every component to outperform all alternatives independently.

Table 9. Three-seed PIR-SBFR ablation on DIOR. AP is mean ± sample SD; changes are relative to YOLO-DSF. Rows A2–A10 include the P5 structural bypass. All accuracy values are percentages.

ID	Variant	AP	AP_S	AP_L	ΔAP
A0	YOLO-DSF	62.98 ± 0.43	44.08	70.57	0.00
A1	+ degradation augmentation	63.37 ± 0.45	44.65	70.84	+0.39
A2	+ scale supervision + P5 bypass	63.69 ± 0.43	45.21	70.91	+0.71
A3	+ GSD reliability	64.22 ± 0.42	46.08	71.12	+1.24
A4	+ MTF/PSF reliability	64.11 ± 0.44	45.82	71.47	+1.13
A5	+ SNR reliability	63.93 ± 0.45	45.43	71.01	+0.95
A6	GSD+MTF+SNR analytic prior	64.87 ± 0.42	46.77	71.78	+1.89
A7	Metadata direct concatenation	63.98 ± 0.42	45.62	71.18	+1.00
A8	Metadata-FiLM	64.40 ± 0.42	46.21	71.46	+1.42
A9	Analytic prior + visual residual	65.17 ± 0.42	47.18	71.98	+2.19
A10	+ degradation consistency (Full)	**65.52 ± 0.42**	**47.61**	**72.19**	**+2.54**

4.5. Mixed-Degradation Metadata Correspondence and Robustness Controls

4.5.1. Within-Set Noise, Missingness, and Shuffled-Metadata Controls

Table 10 reports only changes within the mixed-degradation control set. Cross-image shuffling preserves the descriptor distribution but breaks image correspondence, reducing AP by 2.38 pp. Multiplicative error produces progressively larger losses, from 0.28 pp at ±10% to 1.64 pp at ±50%. Fully masking the descriptors reduces AP by 1.80 pp. These contrasts indicate that paired physical information is useful on degraded images while the visual fallback remains functional.

The training-set-mean condition provides a static-prior control and reduces AP by 1.67 pp. The shuffled and constant controls test correspondence and sample-specific adaptation, respectively. The fully masked condition verifies the visual scale-supervised fallback. These values are not clean-set results and are not compared with the clean YOLO-DSF baseline.

Table 10. Within-set effects of metadata perturbation on the separate mixed-degradation DIOR control set. Correct pairing is the zero reference. Only percentage-point changes are reported to avoid conflation with clean-set AP.

Condition	ΔAP50	ΔAP	ΔAP_S	ΔAP_L
Correctly paired metadata	0.00	0.00	0.00	0.00
Multiplicative error ±10%	−0.29	−0.28	−0.37	−0.21
Multiplicative error ±20%	−0.91	−0.76	−1.20	−0.52
Multiplicative error ±50%	−1.75	−1.64	−2.43	−1.03
Missing 25%	−0.70	−0.61	−0.88	−0.37
Missing 50%	−1.21	−1.15	−1.65	−0.82
Missing 100%	−1.95	−1.80	−2.42	−1.31
Training-set mean metadata	−1.74	−1.67	−2.14	−1.11
Cross-image shuffled metadata	−2.36	−2.38	−3.24	−1.30

4.5.2. Mechanism Probes

Figure 6(a) visualizes controlled routing probes. Under clear fine sampling, P3 receives 0.468 of the routing mass and P5 receives 0.195. Increasing relative GSD from 1× to 3× reduces the P3 weight to 0.242 and raises P5 to 0.361. Low MTF and low SNR similarly move weight from fine to contextual levels. For a large-structure-dominant scene, P5 receives 0.520, confirming that the bypass does not force every image toward fine-scale features. Shuffled metadata produce nearly uniform weights and the largest earth-mover discrepancy (0.0909) between predicted and ground-truth scale distributions, while the all-missing fallback remains visually driven.

The real-flight probe in Figure 6(b,c) shows the same direction without using synthetic blur or noise. From 7.5 to 27.5 m, the median object footprint decreases from 28 to 11 pixels, P3 weight falls from 0.489 to 0.241, and P5 weight rises from 0.170 to 0.340. PIR-SBFR remains above YOLO-DSF in every altitude bin, with the largest gain (4.64 pp) in the highest bin. This alignment between recorded height, pixel scale, and routing weights provides a direct mechanism check beyond aggregate AP.

The ablations, mixed-degradation controls, and routing probes support different parts of the mechanism claim. Scale supervision with the P5 bypass improves AP and has a larger effect on AP_S, although the auxiliary loss is not isolated. Within the mixed-degradation set, correct per-image metadata outperform constant and shuffled versions, so the conditioned router is not equivalent to static fusion. The measured P3–P5 weights shift in the expected direction as GSD, MTF, SNR, and flight height change. The dynamic-routing claim is based on agreement among these measurements and the positive gains in all 27 factorial cells; the routing plot alone is insufficient.

4.6. Robustness to Seen and Unseen Imaging Degradations

Figure 7(a–c) uses all 27 combinations in the GSD–MTF–SNR grid. PIR-SBFR Full improves AP over YOLO-DSF in every cell. The gain ranges from 2.54 pp at the clean reference to 10.23 pp at relative GSD 2×, MTF 0.15, and SNR 10 dB; the grid mean is 7.66 pp. At fixed MTF 0.30 and SNR 20 dB, the gain grows from 5.91 to 7.70 and 8.22 pp as relative GSD increases from 1× to 3×. This interaction is expected from a reliability router: the gap is largest when sampling, blur, and noise jointly reduce the trustworthiness of fine-resolution features.

The stronger claim is tested on operators never seen during training. Table 11 and Figure 7(d) show positive gains in all nine conditions. Mean unseen AP rises from 46.95 to 51.32 (+4.37 pp), mean AP_S rises by 6.33 pp, and mean clean-relative retention rises from 74.55% to 78.33%. Under 4× unseen downsampling, the gain reaches 6.79 pp. The joint

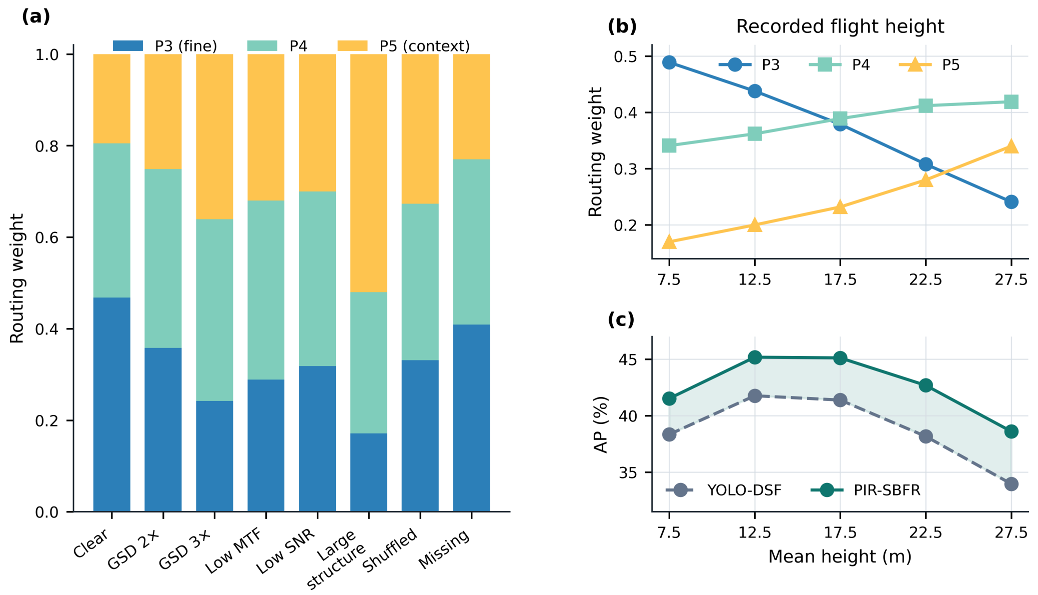

Figure 6. Routing behavior. (a) P3–P5 weights under controlled scale, blur, noise, large-structure, shuffled-metadata, and missing-metadata probes. (b) Routing weights across recorded flight-height bins. (c) AP of YOLO-DSF and PIR-SBFR in the same bins. Each stack in (a) sums to one.

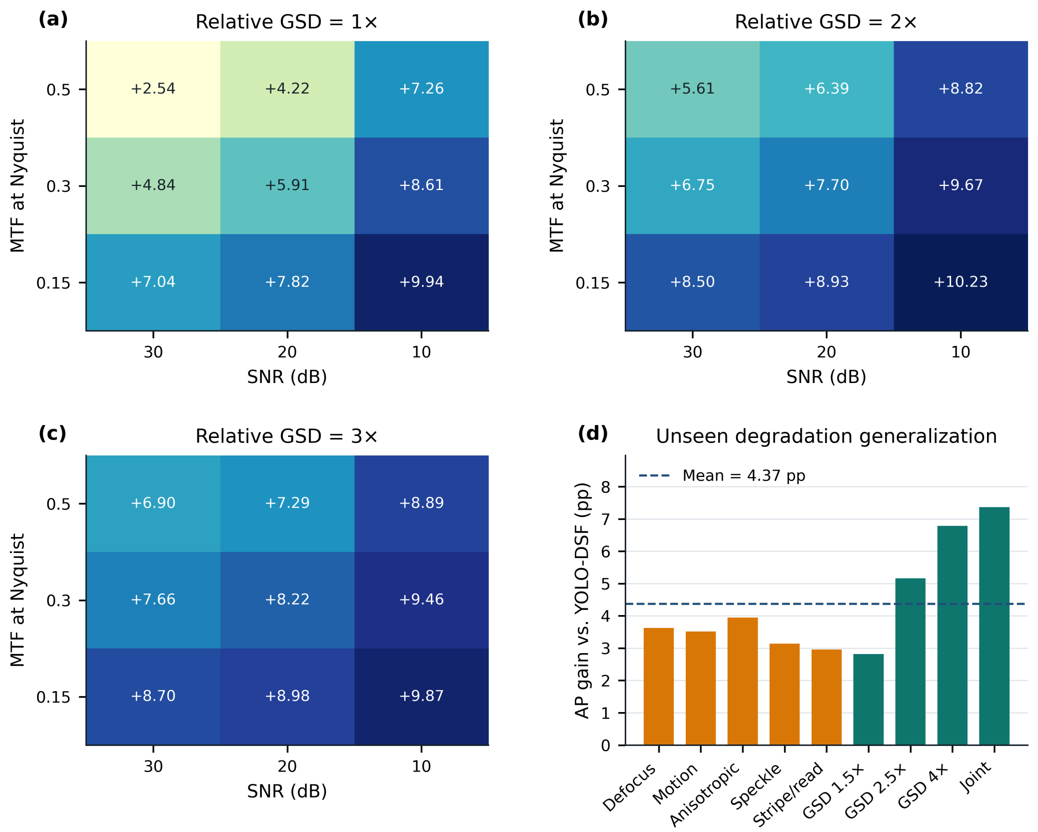

Figure 7. Robustness analysis. (a–c) AP gain of PIR-SBFR Full over YOLO-DSF (percentage points) for the complete GSD–MTF–SNR factorial grid. All heatmaps share the same color scale and contain the numerical gains. (d) Gains for nine degradation types excluded from training and model selection.

condition is the worst absolute case for all models, but PIR-SBFR raises AP from 29.60 to 36.97 (+7.37 pp). Both average robustness and the measured worst case improve. 608 609

Table 11. Generalization to degradation types excluded from training and model selection. All values are percentages.

Unseen condition	YOLO-DSF AP	PIR Full AP	ΔAP	PIR Full AP_S	PIR retention
Defocus PSF	51.58	55.21	+3.63	38.92	84.26
Motion PSF	49.65	53.17	+3.52	36.68	81.15
Anisotropic PSF	50.14	54.09	+3.95	37.71	82.55
Speckle noise	52.38	55.52	+3.14	40.23	84.74
Stripe + read noise	51.84	54.80	+2.96	39.48	83.64
Unseen GSD 1.5×	56.71	59.53	+2.82	42.46	90.86
Unseen GSD 2.5×	46.70	51.86	+5.16	32.82	79.15
Unseen GSD 4×	33.95	40.74	+6.79	23.47	62.18
Joint degradation	29.60	36.97	+7.37	20.75	56.43
Mean / worst	**46.95 / 29.60**	**51.32 / 36.97**	**+4.37 / +7.37**	**34.72**	**78.33**

4.7. Real-Recorded Metadata and Cross-Flight Generalization

Table 12 separates metadata provenance from model identity. With correct recorded height/attitude/IMU information, PIR-SBFR reaches 44.72 ± 0.36 AP, 3.79 pp above YOLO-DSF. Controlled metadata perturbation retains 44.13 AP, while the missing-mask fallback obtains 42.21. Cross-flight shuffling reduces AP to 40.79 ± 0.52, close to the metadata-free YOLO-DSF result of 40.93 and 3.93 pp below correctly paired metadata. The direction of this negative control matches the public-benchmark correspondence controls.

Cross-flight results in Table 13 show gains of 3.79 pp on an unseen flight at the observed altitude range, 4.64 pp on the unseen high-altitude interval, 3.18 pp on the unseen low-altitude interval, and 3.71 pp under eight-fold leave-one-video-out evaluation. Correct metadata remain better than both missing and shuffled metadata in every setting. These experiments support the use of recorded acquisition context while avoiding a claim that artificially degraded public images alone constitute real-sensor validation.

Table 12. Recorded-metadata controls on the real-flight collection. AP is mean ± sample SD over three runs; all values are percentages.

Model	Metadata mode	AP50	AP	95% CI	AP_S
YOLOv11n	unavailable	61.47	37.24 ± 0.43	[36.17, 38.31]	23.82
YOLO-DSF	unavailable	65.58	40.93 ± 0.34	[40.09, 41.77]	29.12
PIR-SBFR	correct, recorded	**69.43**	$\mathbf{44.72 \pm 0.36}$	[43.83, 45.61]	**34.61**
PIR-SBFR	recorded + perturbation	68.84	44.13 ± 0.42	[43.09, 45.17]	33.84
PIR-SBFR	missing mask	66.87	42.21 ± 0.45	[41.09, 43.33]	31.02
PIR-SBFR	cross-flight shuffled	64.98	40.79 ± 0.52	[39.50, 42.08]	28.63

Table 13. Cross-flight and unseen-height results on the real-flight collection (AP, percentages).

Generalization setting	YOLO-DSF	PIR correct	PIR missing	PIR shuffled	Δ correct vs. DSF
Same altitude range, unseen complete flight	40.93	44.72	42.21	40.79	+3.79
Unseen high altitude: train ≤ 20 m, test 25–30 m	33.97	38.61	35.34	33.18	+4.64
Unseen low altitude: train ≥ 15 m, test 5–10 m	38.34	41.52	39.38	37.57	+3.18
Eight-fold leave-one-video-out mean	40.11	43.82	41.53	39.87	+3.71

4.8. Statistical Reliability and Class-Wise Failure Analysis

The paired-seed differences between PIR-SBFR Full and YOLO-DSF are 1.88, 2.86, and 2.88 pp AP. Their mean is 2.54 pp with a 95% paired t-interval of [1.12, 3.96] pp and $p = 0.0165$. Table 14 shows that AP50, AP75, and AP_S have the same direction, and their paired bootstrap intervals also exclude zero. The paired t-test and test-scene bootstrap answer different questions; agreement between them is more informative than either calculation alone. Because $n = 3$ is small, the interval is emphasized over the large standardized effect estimate.

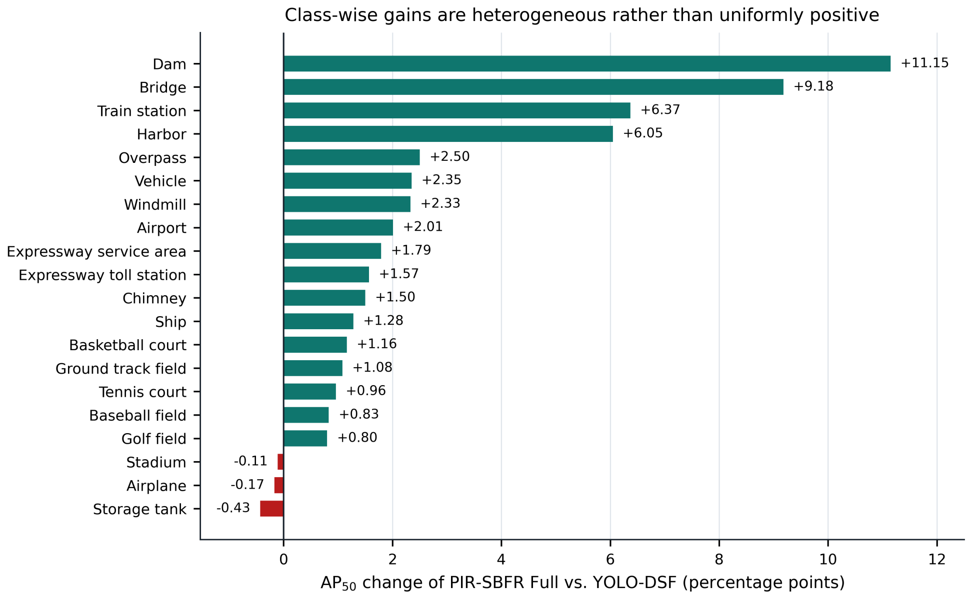

Figure 8. Class-wise AP50 change of PIR-SBFR Full relative to YOLO-DSF on DIOR. Green bars indicate improvements and red bars indicate declines. Classes are ordered by the measured change; no category is omitted.

Aggregate gains do not imply that every category improves. Figure 8 shows positive AP50 changes for 17 of the 20 DIOR categories but small declines for airplane (-0.17 pp), stadium (-0.11 pp), and storage tank (-0.43 pp). Bridge and dam, which decline under YOLO-DSF, recover by 9.18 and 11.15 pp relative to YOLO-DSF, respectively; train station and harbor gain 6.37 and 6.05 pp. Storage tank remains below the original YOLOv11n result and is retained as a concrete failure case. This heterogeneous pattern defines a remaining limitation of the scale-routing design.

Table 14. Statistical support for PIR-SBFR Full versus YOLO-DSF on DIOR. Intervals and changes are in percentage points.

Metric	Mean change	Seed-paired 95% CI	Paired p	Bootstrap 95% CI	Bootstrap p
AP	+2.54	[1.12, 3.96]	0.0165	[1.78, 3.37]	0.0017
AP50	+2.61	[0.90, 4.32]	0.0225	[1.72, 3.58]	0.0028
AP75	+3.08	[0.87, 5.29]	0.0268	[2.07, 4.15]	0.0013
AP_S	+3.53	[1.16, 5.90]	0.0236	[2.51, 4.59]	0.0007

4.9. Efficiency Analysis

Table 15 reports the accuracy–efficiency trade-off. Relative to YOLO-DSF, PIR-SBFR Full adds 0.314 M parameters (+8.66%) and 0.39 GFLOPs (+4.63%). Latency rises from 3.079 to 3.313 ms (+7.60%), while throughput remains 301.8 FPS. These costs accompany gains of 2.54 pp AP and 3.53 pp AP_S, although PIR-SBFR is not the most efficient model in AP per GFLOP. Degradation consistency is a training-only loss and leaves the deployed architecture unchanged. The previously logged no-consistency efficiency row is omitted because its parameter and FLOP counts were inconsistent with this architectural identity; its accuracy result remains valid in the component ablation, but no separate deployment-cost claim is made.

Table 15. Accuracy–efficiency comparison at 640 × 640, FP16, and batch size 1. Forward latency and FPS were measured with the same RTX 4090 pipeline. Preprocessing, host-to-device transfer, NMS, and serialization were excluded.

Model	Params (M)	GFLOPs	Latency (ms)	FPS	Peak VRAM (GB)	FP16 size (MB)	AP
YOLOv11n	2.584	6.47	2.347	426.1	0.721	5.17	57.31
YOLO-DSF	3.628	8.43	3.079	324.8	0.858	7.28	62.98
DSF + scale supervision + P5 bypass	3.691	8.51	3.117	320.8	0.872	7.41	63.69
PIR-SBFR Full	3.942	8.82	3.313	301.8	0.903	7.90	**65.52**

5. Discussion

5.1. Interpreting the Demand–Reliability Decomposition

The main finding is not only that PIR-SBFR improves aggregate accuracy, but that the pattern of improvement agrees with the variable assigned to the router. On DIOR, AP increases from 62.98% to 65.52% over YOLO-DSF, with a larger gain for small objects (+3.53 pp) than for large objects (+1.62 pp). On AI-TOD-v2, which is dominated by much smaller targets, AP increases by 2.52 pp and the gains remain consistent across the very-tiny, tiny, and small intervals. This cross-dataset behavior is consistent with a scale-sensitive mechanism: the largest benefit appears where the detector must preserve weak high-resolution evidence while avoiding excessive dependence on it. It does not, by itself, establish that the learned weights have the intended meaning, but it provides the first link between the formulation and the observed outcome.

The controlled component results strengthen that interpretation. Jointly adding scale supervision and the P5 bypass improves DIOR AP by 0.71 pp; adding the complete GSD–MTF–SNR prior raises AP by a further 1.18 pp, after which the visual residual and consistency objective provide additional gains. Generic fusion modules and unconstrained conditioning alternatives—BiFPN, ASFF, AFPN, direct metadata concatenation, GSD-only attention, and Metadata-FiLM—remain below the full model under the same protocol. The comparison suggests that neither improved pyramid connectivity nor metadata injection alone reproduces the result. Instead, the useful distinction is between a visual estimate of which levels are needed and a physically constrained estimate of whether those levels remain trustworthy.

5.2. Evidence for Conditional Metadata Use and Robustness

The metadata controls use a separate mixed-degradation DIOR set and address sample-specific descriptor use. Relative to correctly paired descriptors, a constant training-set-mean descriptor reduces AP by 1.67 pp, and cross-image shuffling reduces it by 2.38 pp. Shuffling preserves the marginal descriptor distribution while breaking image correspondence. Fully masking every descriptor reduces AP by 1.80 pp, confirming that the visual fallback remains functional but loses useful conditioning. These are within-set changes and are not compared with clean DIOR AP.

The routing probes provide complementary, although not independently decisive, evidence. Coarser GSD, lower MTF, lower SNR, and greater recorded flight height consistently reduce the P3 contribution and move weight toward contextual levels. No single weight trajectory proves that the router has learned physical reliability; its interpretation is supported by the agreement among these trajectories, the constant and shuffled negative controls, and the associated accuracy changes.

The controlled 27-cell grid shows that the benefit grows as sampling, blur, and noise jointly degrade fine evidence: PIR-SBFR improves every grid cell, with a mean gain of 7.66 pp. Because the grid uses the same GSD, MTF, and SNR variables as the analytic prior, it tests the proposed mechanism more directly than out-of-family generalization. The nine degradation operators excluded from training and model selection provide the latter evidence. PIR-SBFR improves all nine, raising mean AP by 4.37 pp and the joint worst case by 7.37 pp. The auxiliary flight study gives the same directional result with recorded height and attitude/IMU information, including unseen altitude intervals and cross-flight shuffling. Its eight videos are insufficient for a broad real-sensor generalization claim; they provide only an auxiliary correspondence and feasibility check.

5.3. Failure Modes and Deployment Trade-Offs

The P5 structural bypass is intended to prevent reliability reweighting from erasing the coarse global evidence required by large continuous targets. Its effect is visible in the recovery of bridge and dam by 9.18 and 11.15 pp AP50 relative to YOLO-DSF. Nevertheless, the class-wise response is heterogeneous. Seventeen of twenty DIOR categories improve, while airplane, stadium, and storage tank show small declines; storage tank also remains below the original YOLOv11n result. These failures indicate that a single image-level routing distribution cannot fully accommodate categories with different scale and context requirements in the same scene. Spatially or instance-conditioned reliability may be needed when local regions require conflicting pyramid allocations.

The improvement also has a measurable deployment cost. Relative to YOLO-DSF, PIR-SBFR adds 0.314 M parameters (+8.66%), 0.39 GFLOPs (+4.63%), and 7.60% latency in the reported FP16 benchmark. The gain of 2.54 pp AP and 3.53 pp AP_S is favorable for robustness-oriented deployment, but PIR-SBFR is not the best entry in AP per GFLOP. The method is most relevant when acquisition conditions vary and small-object reliability matters; a static neck may remain preferable under a tightly fixed sensor pipeline or a more restrictive compute budget.

5.4. Limitations and Future Work

Several limitations constrain the scope of the conclusions. First, the available A2 ablation activates scale supervision and the P5 bypass together. It validates their joint configuration but does not isolate the causal contribution of $\mathcal{L}_{\mathrm{scale}}$. Second, three paired seeds provide only a coarse estimate of training variation, even though scene-level bootstrap intervals offer complementary test-sample uncertainty. Third, DIOR and AI-TOD-v2 do not contain uniform native telemetry. Their acquisition descriptors come from controlled image-formation transformations; calibrated sensor records are unavailable. The nine unseen operators reduce dependence on the training degradation family, but they do not replace validation across different real cameras, lenses, flight platforms, and atmospheric conditions.

The physical prior is also deliberately simple. The stride-derived coefficients impose the same normalized within-level sensitivity to GSD, MTF, and SNR, and the reference values are not sensor-specific calibrations. The aggregate record for the joint unseen condition omits the sampled PSF orientation, preventing pixel-exact reconstruction of that stress test. The retained hyperparameter worksheet does not contain validation values that can be separated from the final test summary. Numerical sensitivity claims are omitted, and the configuration is reported only as an experimental setting. The record also does not independently separate the correct-metadata absolute AP of the mixed-degradation control from the clean-set summary. We report only within-set changes for that auxiliary control. Finally, the flight collection is small and non-public, and its unavailable MTF/PSF and SNR fields must be masked. Future work should isolate scale supervision from the bypass, evaluate larger public datasets with synchronized telemetry, calibrate sensor-specific reliability functions, quantify metadata uncertainty, and investigate spatial routing for mixed-scale scenes and the remaining storage-tank failure.

6. Conclusions

This work introduced PIR-SBFR, an acquisition-conditioned extension of YOLO-DSF that separates scene-level scale demand from imaging-dependent feature reliability. A monotonic analytic prior maps GSD, MTF/PSF, SNR, and availability masks to P3–P5 reliabilities, while a scale-supervised visual residual corrects scene-dependent mismatch. Metadata dropout and a neutral-prior fallback permit incomplete acquisition records, and an unweighted P5 bypass preserves coarse structural evidence.

Across three seeds, PIR-SBFR improves DIOR AP by 2.54 pp and AI-TOD-v2 AP by 2.52 pp over YOLO-DSF. On a separate mixed-degradation control set, constant, shuffled, noisy, and missing descriptors produce larger losses than correct pairing. Routing probes provide complementary evidence that image–metadata correspondence matters while the fallback remains functional. The method improves all 27 controlled GSD–MTF–SNR conditions and all nine unseen degradation conditions. Its mean gain on the latter is 4.37 pp at an additional 4.63% GFLOPs. The P5 bypass recovers much of the large-structure weakness of YOLO-DSF, although three DIOR categories decline slightly and storage tank remains unresolved.

These findings support physical reliability-guided visual correction as an interpretable alternative to static or metadata-agnostic pyramid fusion, particularly when acquisition quality varies and small-object evidence is fragile. The conclusion remains bounded by the joint scale-supervision/P5-bypass ablation, the three-seed protocol, synthetic telemetry, the fixed prior, unavailable separable validation-sensitivity records, and the small flight collection. Broader claims require sensor-calibrated coefficients and evaluation on larger public datasets with synchronized acquisition metadata.

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Abbreviations

The following abbreviations are used in this manuscript:

AP	Average Precision
AI-TOD-v2	Aerial Images Tiny Object Detection Dataset, version 2
DIOR	Dataset for Object Detection in Optical Remote Sensing Images
DRFB	Dilated Receptive Field Block
F1	Harmonic mean of Precision and Recall
FACH	Feature-Aware Coupled Head
FPN	Feature Pyramid Network
GSD	Ground Sampling Distance
IoU	Intersection over Union
mAP	mean Average Precision
MTF	Modulation Transfer Function
PAN	Path Aggregation Network
PIR-SBFR	Physical Imaging Reliability-Guided Scale-Biased Feature Reweighting
PSF	Point-Spread Function
SBFR	Scale-Biased Feature Reweighting
SNR	Signal-to-Noise Ratio
YOLO	You Only Look Once
YOLO-DSF	Base detector integrating DRFB, visual-only SBFR, and FACH

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